9 November 1976

Dear Luc,

Pencils give the conjecture from my letter to Katz in the general case.

Let:

X	be smooth and projective	$\mathcal{F}$ a smooth sheaf of rank 1 on $U = X - D$,
D	a smooth divisor	ramified along D, locally defined by
$D' \subset D$	a smooth divisor in D	$\psi\colon \mathbf{F}_p \longrightarrow \overline{\mathbf{Q}}_\ell^*$ and a covering.

of Artin–Schreier type, ramifying as follows: for $x = 0$ a local equation for D, and $x = y = 0$ for D',

$$T^p - T = \frac{y}{x^d}, \qquad \text{with } (d,p) = 1.$$

Let $\mathcal{F}'$ be the virtual sheaf

$$\mathcal{F}' = j_!\mathcal{F} - j_!\overline{\mathbf{Q}}_\ell + d_!(\overline{\mathbf{Q}}_\ell)_{D-D'}.$$

Then

$$\chi(\mathcal{F}') = 0.$$

Proof by induction. Take a general Lefschetz pencil.

a) χ' does not change after passing to the blow-up associated with the pencil [using the vanishing of χ' for the intersection of X with the axis: this is known by induction].

b) χ of the general hyperplane section is 0 [by induction].

c) An accident occurs only when the section becomes tangent to X, D, or D'. Elsewhere one has in fact a product situation [locally upstairs, $(X, \mathcal{F})$ is the product of the section by the parameter of the section].

d) For a general pencil of high degree, each of these accidents contributes:

$$\chi = \alpha(\#\,\text{tg to } X) + \beta(\#\,\text{tg to } D) + \gamma(\#\,\text{tg to } D').$$

Now vary the degree of the pencil.

Lemma. Let $X \subset \mathbf{P}$ be smooth and projective. The class of X in the n-th Segre embedding (by $\mathcal{O}(n)$) is a polynomial in n, of degree equal to $\dim(X)$.

An easy calculation shows that the coefficient of $n^{\dim X}$ is

$$\deg_X \mathcal{O}(1) = [c_1\mathcal{O}(1)]^{\dim X}.$$

(relativized to pencils, this extends from X to a hyperplane section…)

It follows that, varying the degree,

$$\chi = \alpha(\text{pol } d^\circ \dim X) + \beta(\text{pol } d^\circ(\dim X - 1)) + \gamma(\text{pol } d^\circ(\dim X - 2)).$$

Hence, if $\dim X \geq 3$, one has $\alpha = \beta = \gamma = 0$ and $\chi = 0$.

The case $\dim X \leq 2$ having already been treated, this completes the proof.

Best regards,

P. Deligne